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MENTOR COPY

Module 6 · Production-Grade Engineering

DAY 18 of 25

# Error Handling, Sub-Workflows & Reliability

"Production mein workflow ka fail hona = client ka gussa."

⚡ AAJ HUM YEH BANAYENGE

## Global Error Workflow → Telegram + Log

Ek baar banao, sab workflows par lagao: koi bhi workflow fail ho to 2 second mein Telegram par alert (workflow, node, error, execution link) aur error log sheet mein entry.

Error Trigger | Execute Workflow | Stop and Error

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

# 🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

**💡 Yeh lesson kyun zaroori hai?**

Demo aur production mein farak: production workflow fail hone par bhi graceful rehta hai aur aapko turant batata hai.

## 🎯 Aaj ke baad aap yeh kar paoge

**01 Error Trigger & global error workflows**
🌍 Station master ko har problem ki khabar.
⚙️ Error Trigger workflow → Settings mein har workflow ka Error Workflow set.

**02 Retry on fail, continue on error, fallback paths**
🌍 Ghanti nahi baji? Phir bajao.
⚙️ Node Settings: Retry on Fail (3, 5s), On Error: Continue (output).

**03 Stop and Error node — custom failures**
🌍 Teacher ne bola "galat hai, ruko".
⚙️ IF amount < 0 → Stop and Error "Invalid amount".

**04 Sub-workflows: Execute Workflow & reusable modules**
🌍 Lego ka ek block baar-baar use.
⚙️ Execute Workflow (sub-workflow) with inputs.

**05 Idempotency & duplicate prevention**
🌍 Attendance do baar nahi lagti.
⚙️ Sheet/DB mein order_id check → mila to skip.

**06 Logging & alerting (Slack/Telegram)**
🌍 CCTV + alarm.
⚙️ Slack/Telegram alert: workflow, node, error, execution URL.

## ⏱ Class ka plan (2 ghante)

| 10m | 35m | 45m | 20m | 10m |
|---|---|---|---|---|

`0:00-0:10` **Recap**

*🪄 Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.*

👩‍🏫 2 students se homework screen share karwao. Galti wala homework bhi dikhao — sabse zyada seekh wahin se aati hai.

`0:10-0:45` **Concept: Reliability Engineering**

*🪄 Cycle ki chain utar gayi to tum ro ke nahi baithte — chadhate ho, nahi chadhi to papa ko bulate ho.*

👩‍🏫 Pehle real-life example se shuru karo, phir n8n screen dikhao. Har concept ke baad 1 student se apne shabdon mein dobara bulwao.

`0:45-1:30` **Live Build**

Central error handler + alert + retry queue

*🪄 Aaj ki kahani yaad hai — 'Train ka safety system'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.*

👩‍🏫 Har 10 minute par poocho "sabka green tick aaya?". Jaan-boojh ke ek galti live hone do aur use debug karke dikhao.

`1:30-1:50` **Chaos Lab**

Hum workflow todenge, aap bachaoge

*🪄 Practice match — jitna yahan girenge, utna asli match mein kam girenge.*

👩‍🏫 Pehle 5 minute akele try karne do, phir hint do. Jo solve kare use screen share karwao.

1:50-2:00 **Q&A + Homework**

*🪄 Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?*

👩‍🏫 Chat se top 3 doubts live solve karo. Homework ka pehla step class mein hi karwa do, taaki koi blank na baithe.

## 🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

### Error workflow

Har workflow ki settings mein ek "Error workflow" set karo.

🪄 **Analogy:** Fire alarm

CONCEPT 02

### Retry on fail

Node fail ho to X baar, Y second ke gap par retry.

🪄 **Analogy:** Phone na lage to dobara try

CONCEPT 03

### Continue on fail

Error ko output mein bhejo aur alag handle karo.

🪄 **Analogy:** Plan B

CONCEPT 04

### Sub-workflows

Common logic ek jagah — 10 workflows reuse.

🪄 **Analogy:** Lego blocks

### 🔑 Aaj ka main idea

Reliability = workflow fail ho to bhi system sambhal le: retry kare, alternate raasta le, duplicate na kare, aur turant batade. Production mein yeh sabse zaroori skill hai.

- API down → 3 baar retry → phir Telegram alert
- Duplicate order webhook → ek hi baar process
- Common "send WhatsApp" sub-workflow sab jagah use

## 📖 Ek kahani se samjho: Train ka safety system

1 🚆 **Train chal rahi hai** = Workflow run

2 ⚠️ **Signal fail ho gaya** = Node error

3 🔁 **Driver ne dobara try kiya** = Retry on fail

4 🛤️ **Doosri line par moda** = Fallback path

5 📟 **Station master ko alert** = Error Trigger → Telegram

🪄 Ek line mein: “Cycle ki chain utar gayi to tum ro ke nahi baithte — chadhate ho, nahi chadhi to papa ko bulate ho.”

# 🗺️ Aaj ka workflow: Global Error Workflow → Telegram + Log

Data left se right chalta hai. Har dabba ek n8n node hai.




**💰 Isse kitna kama sakte ho?**

Har client project mein add-on (error monitoring) — aam taur par **₹3,000 – ₹8,000 add-on**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

## 🛠 Step-by-step build — n8n kholo aur saath chalo

1. ⚡ **Error Trigger**
   Kisi bhi workflow ka failure
2. ✏️ **Edit Fields (Set)**
   workflow name, node, error message, execution URL
3. ✈️ **Telegram**
   Alert: "❌ Lead Engine fail — HubSpot 401"
4. 📊 **Google Sheets**
   Error log
5. 🔀 **IF**
   Retryable error (429/5xx)?
6. 🧩 **Execute Workflow (Sub-workflow)**
   Retry queue workflow call

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

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### 🔑 Setup (import ke baad)

- [ ] Connect: **Telegram bot token (BotFather se)**
- [ ] Connect: **Google Sheets (Google OAuth)**
- [ ] Is workflow ko save karo (Active karne ki zaroorat nahi)
- [ ] Har doosre workflow mein: Settings → Error Workflow → "Day 18 — Global Error Workflow" chuno
- [ ] "Errors" tab headers: workflow, node, error, url, time

### ✅ Test kaise karein

1. Ek test workflow banao jisme HTTP node galat URL par jaaye, uska Error Workflow set karo
2. Use Active karke fail karwao → Telegram alert aaya?

## ⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

**😩 Pehle (manual)**

Client bole "kaam nahi hua" tab pata chalna — **120 min**

120 min

→

**⚡ Ab (n8n)**

2 second mein alert + link — **1 min**

1 min

⏱ Downtime ghante se minute mein. Client ka bharosa bacha.

### ⌨️ Aaj ka key snippet

**Copy karke n8n mein paste karo** — 📋 Select karke copy karo

```
❌ {{ $json.workflow.name }} failed at {{ $json.execution.error.node.name }}
{{ $json.execution.error.message }}
🔗 {{ $json.execution.url }}
```

### 📦 Sample data — input se output tak

**📥 Input** — 📋 Select karke copy karo

```
{
  "execution": {
    "id": "2231",
    "url": "https://n8n.you.com/execution/2231",
    "error": {
      "message": "401 Unauthorized",
      "node": {
        "name": "HubSpot"
      }
    }
  },
  "workflow": {
    "name": "Lead Engine"
  }
}
```

**📤 Output** — 📋 Select karke copy karo

```
{
  "alerted": true,
  "logged": true,
  "retry": false
}
```

# 💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

### 💬 Build it with ChatGPT / Claude

📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Ek baar banao, sab workflows par lagao: koi bhi workflow fail ho to 2 second mein Telegram par alert (workflow, node, error, execution link) aur error log sheet mein entry.

Flow: Error Trigger (Koi workflow fail) → Edit Fields (Details nikaalo) → Telegram (Alert) → Google Sheets (Error log)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

### 💬 Debug helper prompt

📋 Select karke copy karo

```
I'm building the n8n workflow "Global Error Workflow → Telegram + Log" (nodes: On Any Error, Error Details, Telegram Alert, Error Log).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

## 📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo ( `Ctrl` + `A` , `Ctrl` + `C` ).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

**⚡ Workflow JSON**
4 nodes · ready to import
`adifyskills.com/workflows/day-18-global-error-alert.json`

**🌐 Lesson page**
Copy buttons, quiz aur video links
`adifyskills.com/day/18`

# Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

| Agar yeh ho raha hai… | To yeh karo |
|---|---|
| **Node par laal ⚠️ triangle?** | Account connect nahi hua. Node kholo → Credential → "Telegram bot token (BotFather se)" connect karo. |
| **Node par laal ⚠️ triangle?** | Account connect nahi hua. Node kholo → Credential → "Google Sheets (Google OAuth)" connect karo. |
| **Time par apne aap nahi chala?** | "Active" toggle on hai? Settings mein timezone Asia/Kolkata check karo. |
| **"Sheet not found" / column khaali?** | Sheet ID aur tab ka naam (capital/small bhi) check karo. Pehli row mein headers hone chahiye. |

### ⚠️ Yeh galtiyan sabse common hain

- ! Error workflow banaya par workflows ki settings mein set nahi kiya
- ! Har error par retry (401 retry karne se theek nahi hota)
- ! Alerts itne zyada ki koi dekhe hi nahi

### ⭐ Pro tips — time bachane wali tricks

- ★ Idempotency key (order ID) se duplicate side-effects roko
- ★ Sub-workflow inputs ko clearly define karo
- ★ Weekly error report — trends dekho

**Phir bhi atke? Koi baat nahi 🙂**
Screenshot + error message ke saath WhatsApp par poochho →

# Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

### 🛠 Live lab

Global error workflow: kisi bhi workflow fail par Telegram alert with error details + execution link.

### 📝 Homework

Apne purane 3 workflows mein proper error handling + sub-workflows add karo.

- ☐ Homework ho gaya

**BEGINNER**
Error workflow → Telegram ☐

**INTERMEDIATE**
Retry on fail + fallback path ☐

**ADVANCED**
Dead-letter queue in Postgres + replay ☐

### 🧩 Aaj ka challenge — pehle khud try karo

**🎯 Same workflow 10 minute mein 3 baar fail ho to alert mein "🔥 Repeated failure" likha aaye.**

**💡 Hint 1**
Pichhle errors Error Log sheet mein hain — unhe padh sakte ho.

**💡 Hint 2**
Sheets read → Code mein last 10 min ke same workflow errors gino.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

## 🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

**Q1. Global failure handler ka trigger?**

- (A) Manual
- (B) Form
- (C) Error Trigger
- (D) Webhook

**Q2. Kaunsa error retry karne layak?**

- (A) 401
- (B) 400 validation
- (C) 404
- (D) 429 / 503

**Q3. Reusable logic ke liye?**

- (A) Sub-workflow
- (B) Copy-paste
- (C) Sticky note
- (D) Pin

**Q4. Custom failure throw karne ka node?**

- (A) Sort
- (B) Stop and Error
- (C) NoOp
- (D) Merge

## 📖 Aaj ke naye shabd

**Workflow**
Steps ki ek chain jo apne aap chalti hai — jaise recipe ke steps.

**Node**
Workflow ka ek dabba/step. Har node ek kaam karta hai (email bhejna, data laana...).

**Trigger**
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

**Execution**
Workflow ka ek baar chalna. Har run ka record Executions tab mein dikhta hai.

**Retry**
Fail hone par apne aap dobara try karna.

**Sub-Workflow**
Ek workflow jo doosre workflow ke andar se chalaya jaata hai — reusable block.

**Idempotency**
Ek hi kaam do baar aaye to bhi result ek hi baar ho (duplicate se bachna).

# 👨‍🏫 Mentor notes — class kaise lein

Yeh page sirf aapki copy mein hai. Students ki PDF mein nahi jaata.

- [ ] Chaos Lab: hum jaan-boojh kar credentials todenge
- [ ] Students apne purane workflows "harden" karein
- [ ] Error ko "kab retry, kab alert" matrix se samjhao

## 🗣️ Har block ke liye mentor tip

**10m — Recap**

2 students se homework screen share karwao. Galti wala homework bhi dikhao — sabse zyada seekh wahin se aati hai.

Class se poochho: Kal wale workflow mein sabse mushkil step kaunsa laga? · Homework mein kahan error aaya aur kaise theek kiya? · Kal ka ek node naam batao aur woh kya karta hai?

**35m — Concept: Reliability Engineering**

Pehle real-life example se shuru karo, phir n8n screen dikhao. Har concept ke baad 1 student se apne shabdon mein dobara bulwao.

Class se poochho: Retry aur Continue on Fail mein farak? · Idempotency ka ek example? · Error alert mein kya-kya hona chahiye?

**45m — Live Build**

Har 10 minute par poocho "sabka green tick aaya?". Jaan-boojh ke ek galti live hone do aur use debug karke dikhao.

Class se poochho: Is node ka output agle node mein kaise ja raha hai? · Agar yeh node fail ho to kya hoga? · Isko apne kaam/business mein kahan use kar sakte ho?

**20m — Chaos Lab**

Pehle 5 minute akele try karne do, phir hint do. Jo solve kare use screen share karwao.

Class se poochho: Global failure handler ka trigger? · Kaunsa error retry karne layak? · Reusable logic ke liye?

**10m — Q&A + Homework**

Chat se top 3 doubts live solve karo. Homework ka pehla step class mein hi karwa do, taaki koi blank na baithe.

Class se poochho: Aaj ka sabse useful node kaunsa laga? · Homework mein pehla step kya hoga? · Aaj ki class 1 line mein summarize karo.

## ✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 19

**Self-Hosting n8n: Docker, VPS & Domains**

"Apna khud ka n8n server — unlimited workflows."

☐ **Day 18 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

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# Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

**Q1 → C**
Error Trigger

**Q2 → D**
429 / 503

**Q3 → A**
Sub-workflow

**Q4 → B**
Stop and Error

### ✅ Challenge ka solution

Error Details → Google Sheets (read Errors) → Code: count rows where workflow same and time > now-10min → Telegram text mein count >= 3 ho to 🔥 prefix.

## 📚 Official docs

- Error handling ↗
- Sub-workflows ↗
- Execute Workflow ↗
- Error Trigger ↗
- Workflow settings ↗

## ▶ Videos

- n8n error handling workflow ↗
- n8n sub workflows tutorial ↗

**Doubt poochho**
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